

Advanced Aircraft Design & Simulation

From Concept to Complete 3D Aircraft Modeling & Simulation

Duration: 12 Weeks **Software:** CATIA V5 | ANSYS Workbench **Fee:** Rs. 6,500 (50% OFF)

Weekly Curriculum

Week 1 — Aircraft Design Foundations

- Aircraft classifications & components
- Aerodynamics & flight mechanics
- NACA series & airfoils
- Structures, materials, propulsion

Week 2 — Conceptual Design & CATIA

- CATIA V5 interface & workbenches
- Sketch-based modeling
- Aircraft part design

Week 3 — Assembly & Drafting

- Constraint management
- Motion simulation
- 2D drawings & GD&T;
- BOM

Week 4 — Surface Design (Pro Level)

- Generative Shape Design
- Lofting, Sweep, Blend
- Curvature analysis

Week 5 — Complete Aircraft Surface

- Fuselage, wing, tail surfaces
- Landing gear
- Final OML model

Weeks 6-7 — Wireframe & Final Assembly

- Wireframe elements
- Winglet integration
- Final assembly verification

Week 8 — ANSYS Simulation Intro

- Geometry import
- Meshing fundamentals
- Aerodynamic & structural analysis

Weeks 9-12 — Capstone Aircraft Project

- Complete aircraft design
- Aerodynamic & structural validation
- Final report & presentation

What You Will Receive

- Complete 3D Aircraft Model (CATIA)
- ANSYS Simulation Files
- Technical Project Report
- Engineering Presentation
- Industry-Recognized Certificate

Contact: info@vedricxglobal.com | +91 8439120370